

Changes from `paper3_main_v8` to `v12`

Consolidated record for the co-author. Covers four revision steps:

step	date	nature
v8 → v9	16 Sep 2026	every number regenerated from a single consistent pipeline
v9 → v10	17 Sep 2026	figures consolidated into panels; per-text figures moved to an appendix
v10 → v11	18 Sep 2026	statistics (permutation tests, Spearman), notation collisions, text corrections
v11 → v12	19 Sep 2026	Physica A typesetting; split into main text + Supplementary Material

Every number is produced by the released code from the tokenised texts with fixed seeds, and is reproduced by `scripts/paper_numbers.py data/corpus`, which writes `results/paper_numbers.md`.

Status column: `done` = already in the `.tex`; `pending` = agreed but not yet applied. Nothing marked `pending` should be taken as final. Statuses were re-verified against the source on 19 Sep: of the nine items listed as pending on 18 Sep, all nine are now applied.

Integrity of v12, checked mechanically against v8 and against `SI.tex`: no duplicate labels within either file and none shared between them; no dangling `\ref` in either direction; 15 references resolve from the main text into the SI through `\xr`; all environments balanced; every `\cite` key present in the bibliography. Every label present in v8 and absent from v12 is an intended consolidation, itemised in §2 and §4.3 — nothing else was lost.

0. What matters most, in five points

- The pipeline is reproducible.** The four Pearson correlations regenerate exactly (0.702, 0.295, 0.511, 0.571), verified against `scipy` by an assertion inside `paper_numbers.py`. Everything else in this document rests on that.
- The FGN surrogate is a permutation of the original text.** The rank-to-word mapping reproduces the empirical rank histogram exactly, so all three sequences — original, word-shuffled, FGN — share the same token multiset and therefore the same embedding covariance, PCA eigensystem and projected marginals, *exactly*. They differ only in the ordering. This is stronger than what the paper currently claims, and it exposed three stale passages that survived from v8 (§3.6).
- The significance tests changed, and two marks moved.** The parametric *t*-test on *r* assumes bivariate normality; the eigenvalue spectrum is strongly skewed, so it was

replaced by a permutation test over the pairing of the two spectra. *War and Peace* drops from (***) to (**), *Ulysses* from (**) to (*). The substantive conclusions are unchanged.

- 4. **Four factual errors were found, all traceable to v8.** The burstiest direction of *Origin of Species* (PC2 → PC6, §3.5); the OOV rate ("5-15%" → under 2%, §3.9); the number of shuffled realisations behind the ACF curve (10 → 1, §3.9); and the claim that the FGN and original covariance structures differ by a few percent (they are identical, §3.6). Three of the four were contradicted by the paper's own tables or captions.
- 5. **The paper is now in submission form.** The manuscript is split into a main text and a Supplementary Material to meet the 30-page limit, with one-directional cross-references (§4.2). In the course of that split a copy-paste error silently deleted two whole sections; they were detected by the dangling references they left behind and restored byte-for-byte (§4.4). The reference list, however, is still not in Physica A's numbered format, and changing `\bibliographystyle` did not fix it (§4.6).

1. v8 → v9 — regenerated numbers

The earlier version mixed results from three separate Colab batches, each of which recomputed PCA and the FGN surrogate independently. The pipeline now computes them once per text, so the burstiness, ACF and DFA tables share the same surrogate (same α_0 , same realisation).

Two consequences worth stating plainly to a referee:

- The FGN surrogate was previously built from a differently tokenised stream (regex `(\w+)`). DFA exponents in the corpus table shift by up to 0.03 (*Ulysses*: 0.794 → 0.763).
- In the previous ACF batch the FGN of every text had been tuned to $\alpha = 0.769$ by a coding slip. The FGN autocorrelation is negligible either way, so the ACF conclusions are unchanged.

1.1 Corpus (status: done)

Alice in Wonderland removed (7,237 filtered tokens — too short for a reliable IET estimate at $q = 0.95$). Corpus is now 16 texts.

quantity	v8	v9+
texts	17	16
raw token range	27,000–565,000	56,000–565,000
filtered token range	12,000–321,000	17,000–236,000
DFA exponent range	[0.61, 0.85]	[0.63, 0.85]

A column α_0 (FGN generator exponent) was added to the corpus table. α_0 is determined only to ± 0.01 by the bisection tolerance and is therefore reported with two decimals throughout.

1.2 Four showcase texts (status: done)

text	α v8 $\rightarrow$ v9	α_0 v8 $\rightarrow$ v9	r v8 $\rightarrow$ v9
War and Peace	0.719 $\rightarrow$ 0.721	0.770 $\rightarrow$ 0.76	0.702 $\rightarrow$ 0.702
Origin of Species	0.769 $\rightarrow$ 0.771	0.804 $\rightarrow$ 0.80	0.294 $\rightarrow$ 0.295
Great Expectations	0.688 $\rightarrow$ 0.689	0.735 $\rightarrow$ 0.74	0.522 $\rightarrow$ 0.511
Ulysses	0.794 $\rightarrow$ 0.763	0.839 $\rightarrow$ 0.80	0.578 $\rightarrow$ 0.571

Mean burstiness over the first 20 components at $q = 0.95$ was previously reported without stating the quantile and with values from a different batch: 1.18 / 1.10 / 1.07 / 1.13 $\rightarrow$ **1.36** (War and Peace), **1.35** (Origin), **1.23** (Ulysses), **1.18** (Great Expectations), against $\bar{B} \simeq 0.97$ -0.98 for both null models in every case.

1.3 Corpus-level burstiness (status: done)

quantity	v8	v9+
$\langle B_k^{\text{orig}} \rangle_{\text{texts}}$	[1.12, 1.37], mean 1.22	[1.16, 1.54], mean 1.26
$\langle B_k^{\text{surr}} \rangle$	[0.972, 0.978]	[0.973, 0.978]
$\langle B_k^{\text{FGN}} \rangle$	[0.970, 0.992]	[0.958, 0.983]
$B^{\text{orig}} > B^{\text{FGN}}$	99.4% of pairs	100% of pairs
$\ B^{\text{FGN}}/B^{\text{surr}} - 1\ < 0.05$	95.9%	95.6%
per-text mean ratio orig/surr	1.248 ± 0.011	1.29 ± 0.03
FGN/surr ratio	$1.0035 \pm 0.0012, t = 2.88, p = 0.011$	$1.001 \pm 0.001, t = 1.1, P = 0.28$

The last row is the most useful change for the argument. In v8 the two null models were *significantly different* from each other, which was awkward to explain away as estimator noise. With the consistent pipeline they are statistically indistinguishable — the cleaner and stronger statement.

The abstract changed accordingly: "exceeds the FGN surrogate in 99.4% of pairs" → "in all pairs".

The excess of the original over the nulls is now described as a $\sim 30\%$ mean excess (was: a consistent $\sim 25\%$ excess). The per-text spread is reported explicitly: $[1.17, 1.39]$ for fifteen texts with *Principia Mathematica* an outlier at 1.73, $sd = 0.13$.

1.4 ACF and DFA (status: done)

quantity	v8	v9+
GE: $\rho_k(1)$ original range	$[0.042, 0.179]$	$[0.049, 0.208]$
GE: i.i.d. band $1.96/\sqrt{M}$	± 0.006	± 0.008
GE: FGN lag-1	< 0.025 , 19/20 inside band	≤ 0.007 , 20/20 inside band
corpus $\bar{\rho}^{\text{orig}}(1)$	0.096 ± 0.044	0.100 ± 0.045
corpus $\bar{\rho}^{\text{FGN}}(1)$	0.003 ± 0.009	0.002 ± 0.005
FGN pairs significant at lag 1	17.2%	10.9%
t -test FGN $\rho(1)$ vs 0	$t = 5.83$	$t = 5.05$
GE: $\bar{\alpha}^{\text{orig}}$	0.637 ± 0.032	0.623 ± 0.034
GE: $\bar{\alpha}^{\text{FGN}}$	0.502 ± 0.015	0.499 ± 0.014
GE: α_k^{orig} range	$[0.575, 0.698]$	$[0.564, 0.693]$

A remark about PC1 lying outside the i.i.d. band ($\rho_1^{\text{FGN}}(1) = 0.023$) was **deleted**: with the consistent pipeline the value is 0.004, inside the band.

The residual FGN excess at lag 1 is now quantified rather than asserted: the significant cases concentrate on PC1 (56% of PC1 pairs against 8% for $k \geq 4$), consistent with the known GloVe artefact from rare transliterated proper nouns.

2. v9 → v10 — figure consolidation (status: done)

No numbers changed in this step. Structural only:

- Four per-text burstiness-spectrum figures → one 4-panel figure (`fig:burst_panel`), panels A-D).
- Four per-text quantile-sweep figures → one 4-panel figure (`fig:sweep_panel`).
- The four "semantic poles" figures were **removed**; the same information is in the PC tables (`tab:wrnpc_pc` and siblings), which are more precise and cost less space.

- Per-text eigenvalue spectra and IET distributions moved to a new appendix, *Additional per-text figures* (`(app:per-text)`), keeping only *War and Peace* in the main text as the worked example.
- Figure paths now follow the corpus filenames (`(figures/<stem>/<stem>_NN_<name>.png)`), so every figure is regenerable by `(scripts/run_text.py)`.
- Dropbox references replaced by a pointer to the code repository.

Net effect: roughly fifteen figures down to the panels plus the appendix.

3. v10 → v11 — statistics, notation, corrections

3.1 Notation: `(p)` was overloaded (status: done)

`(p)` meant both the **event rate** $p = 1 - q$ (Sec. 2.3, and in the robustness section at $p = 0.01$) and the **p-value** of the correlation and t -tests — both small decimals, a few pages apart. The event rate is structural and appears throughout, so the p-value was renamed `(P)`.

Seven occurrences converted (status: done): the four (r, p) pairs in the per-text sections, and three t -test p-values in the corpus sections.

Four further occurrences used `(<)` and `(>)` rather than `(=)`, so the first pass missed them: the `(tab:single_text_summary)` legend, the t -test in Eq. `(ratio_corpus)`, and the corpus ACF t -test. All are now converted. The only remaining lowercase `(p)` in the source is the event rate itself, in the geometric-baseline derivation — which is the intended meaning.

3.2 Three further symbol collisions (status: done)

symbol	conflicting meanings	resolution
α	matching condition in $u_t^{(\alpha)}$ vs DFA exponent (≈ 65 uses)	matching condition $\rightarrow \mathcal{C}$ (6 occurrences)
γ	Zipf exponent $f(r) \sim r^{-\gamma}$ vs diffusion exponent	diffusion exponent $\rightarrow H$, tied explicitly to the DFA α of the stationary series
$\tilde{s}$	retained tokens $(\tilde{s}_1, \dots, \tilde{s}_M)$ vs FGN surrogate word sequence	surrogate $\rightarrow \mathbf{s}^{\text{FGN}}$ (1 occurrence)

H and "Hurst" appear nowhere else in the paper, so this introduces no new collision.

3.3 Significance by permutation, not by the parametric test (status: done)

The $K = 20$ pairs (λ_k, B_k) are not independent draws: $\{\lambda_k\}$ is a deterministic decreasing sequence and both spectra derive from the same trajectory. The t -test on r is therefore not a valid significance test here. It is replaced by a permutation test over the pairing between the

two spectra (10^5 pairings, seed 0), implemented in `scripts/pearsonr_spearman.py` and called from `paper_numbers.py`.

text	r	parametric p	permutation P	mark
War and Peace	0.702	0.0006	0.0055	<code>***</code> → <code>**</code>
Origin of Species	0.295	0.207	0.169	none → none
Great Expectations	0.511	0.021	0.0136	<code>*</code> → <code>*</code>
Ulysses	0.571	0.009	0.0406	<code>**</code> → <code>*</code>

The parametric test was anticonservative by up to an order of magnitude, as expected when the predictor is strongly skewed.

Note for the reader of the table: *Ulysses* has a larger r than *Great Expectations* but a larger P . This is not an error — the permutation null depends on the shape of each text's eigenvalue spectrum.

3.4 Spearman added (status: done)

The eigenvalue spectrum decays steeply, so Pearson is dominated by the leading components. The rank correlation was computed as a robustness check and turns out to carry information of its own:

text	r	ρ	P_ρ
War and Peace	0.702	0.850	$< 10^{-4}$
Origin of Species	0.295	0.460	0.042
Great Expectations	0.511	0.579	0.0085
Ulysses	0.571	0.817	$< 10^{-4}$

$\rho > r$ in all four texts: the association between semantic variance and burstiness is **monotone but not linear**, and Pearson understates it.

`P = 0.0000` in the regenerated output is the floor of 10^5 permutations ($1/100001$) and is reported as $P < 10^{-4}$, not as zero.

3.5 Text correction: the burstiness maximum of Darwin (status: done)

The body text of the *Origin of Species* subsection states that PC2 is the burstiest direction. It is not: the maximum is at **PC6**, $B_6 = 1.900$, against $B_2 = 1.888$. The v10 caption of `fig:burst_panel` already gives the maxima correctly as PC3, PC6, PC3, PC2, so the paper

currently contradicts itself. The body text is a leftover from v9, when the now-removed per-text figure `(fig:darwin_burst)` carried the PC2 claim.

This matters beyond the correction. The subsection used the weak correlation as its argument:

The weak eigenvalue–burstiness correlation, $r = 0.295$ ($p = 0.207$), shows especially clearly that the direction of greatest semantic variance need not be the direction of greatest temporal clustering.

That argument no longer survives contact with Spearman: $\rho = 0.460$, $P = 0.042$ is a significant association. The displacement of the maximum to PC6 says the same thing directly, needs no correlation measure at all, and holds in every text (PC3, PC6, PC3, PC2 — never PC1). The subsection is rewritten around that fact, with the two correlations reported as description rather than as the argument.

3.6 The FGN surrogate is a permutation — and three stale passages (status: done)

Verified directly: `(Counter(tokens_all))` of the FGN surrogate equals that of the original, for every showcase text, and the filtered counts agree to the token at $R = 100, 150$ and 200 . Multiset identity at the unfiltered level makes the result independent of R by construction, not by observation.

text	M	M_{FGN}	ratio	multiset identical
War and Peace	235,933	235,933	1.0000	yes
Origin of Species	58,913	58,913	1.0000	yes
Great Expectations	64,410	64,410	1.0000	yes
Ulysses	101,489	101,489	1.0000	yes

Consequences. By Eq. `(sigma_invariant)`, $\Sigma_{\text{FGN}} = \Sigma_{\text{orig}}$ **exactly**: the original eigenvectors *are* the surrogate's own principal directions. The design can now be stated in one line — *the three sequences are permutations of one another and differ only in the ordering of visits through a fixed semantic geometry* — and the two null models differ in exactly one ingredient, the rank-level scaling.

Three passages assert the opposite and are all v8 leftovers, from when the FGN was tokenised with a different regex and the multisets genuinely differed:

- `(rem:fgn_cov)`: "does not impose covariance invariance as an algebraic constraint ... empirically very close" → the invariance *is* algebraic.
- §Text preprocessing: "increasing R to 150 or 200 ... begins to degrade the alignment between the original and FGN covariance structures" → no such degradation is possible; sentence deleted.

- §FGN surrogate generation: "any residual difference is below 3% in eigenvalue and below 5% in eigenvector angle" → the difference is zero.

A fourth passage becomes redundant rather than wrong: the sentence added earlier today noting that M_{FGN} need not equal M is removed, since the two are equal by construction.

One further consequence, in the paper's favour. The code computes the event threshold once from the original series and applies it to all three (`th = thresholds(an.pca.proj, q)`), whereas §2.3 describes each series being thresholded at its own quantile. The two conventions coincide exactly, because permutations share a marginal distribution — so the event rate is exactly $1 - q$ for all three sequences, and this can now be stated as a consequence rather than assumed.

3.7 Does the frequency filter destroy the rank-level scaling? (status: new result)

A referee can reasonably ask whether the FGN's projected series show $\bar{\alpha} = 0.499$ because rank-level memory does not reach embedding space — the paper's claim — or simply because the top- R filter destroyed that memory before the projection. The question was settled by measuring the DFA exponent of the *filtered* rank sequence for both the original and the surrogate.

text	unfiltered	α (pipeline)	original, filtered	FGN, filtered
War and Peace	0.693	0.721	0.674	0.633
Origin of Species	0.726	0.771	0.694	0.645
Great Expectations	0.672	0.689	0.609	0.567
Ulysses	0.805	0.763	0.772	0.629

The filter does not destroy the scaling. Every filtered value stays well above 0.5; the FGN arrives at the projection stage still carrying rank-level memory of $\alpha \simeq 0.57$ –0.65, and its projections nevertheless show none. The claim is supported, now with a number behind it.

But the filter degrades the surrogate more than the original, in every text, and markedly so for *Ulysses* (−0.176 against −0.033). The matching $\alpha_{\text{FGN}} \simeq \alpha_{\text{orig}}$ is imposed before the filter, so at the level where the comparison is made the null carries somewhat less memory than intended. This is a handicap to the null model, not a licence for it, and is worth stating rather than leaving to be found.

Caveat on these four columns: they were computed with `projection_alpha`, which does not reproduce the pipeline's own `alpha_text` (discrepancies up to 0.045, and of opposite sign for *Ulysses*). The three columns are comparable with one another but should not be printed alongside α and α_0 until recomputed with the estimator used inside `fgn_surrogate`.

3.8 Burstiness beyond the retained $K = 20$ (status: new result)

The paper retains $K = 20$ components. Whether anything changes beyond that was previously asserted without evidence; it has now been measured by refitting the PCA with $K = 50$.

text	cum. var. 20	50	100	mean B 1-20	21-50	min B 21-50
War and Peace	37.8%	57.1%	76.4%	1.358	1.138	1.085
Origin of Species	38.1%	59.0%	78.6%	1.350	1.188	1.095
Great Expectations	38.5%	59.0%	78.7%	1.176	1.112	1.041
Ulysses	35.0%	53.2%	72.7%	1.234	1.120	1.041

Every component up to $K = 50$ stays above the independent-event baseline $\backslash CV_{\text{geom}} = 0.975$ in every text; the worst case, 1.041, is still 6.8% above it. Variance per component decays from 1.9% over the first twenty to 0.65% over components 21-50 and 0.39% over 51-100, and mean burstiness falls by 36-61% of its excess over the baseline.

So the qualitative picture does hold, and the honest formulation states both halves: burstiness persists well beyond the retained range while decreasing in magnitude. $K = 20$ is a choice about interpretability, not a claim that nothing happens further out.

The refit reproduces the stored projections to 10^{-6} , which is the float32 precision floor of the GloVe vectors — not a sign flip, which would have produced an error of order unity and would have changed B_k , since the threshold is applied to the upper tail only.

3.9 Errors in the pipeline description (status: done)

Four statements in the methodology do not describe the code that produced the results.

- **OOV rate.** "The OOV rate after filtering is typically 5-15% for Victorian English texts" — the paper's own summary table reports 1.6%, 0.4%, 1.3% and 0.7%, confirmed independently from the pre- and post-lookup token counts (1.58 / 0.44 / 1.34 / 0.73%). Wrong by an order of magnitude, and contradicted fifteen pages later.
- **Number of shuffled realisations.** The code uses three different conventions — 20 realisations for the burstiness spectrum (`(n_surr = 20)`), a single realisation for the ACF (`(a.pca.proj[a.perms[0]])`, so labelled in the source), and 10 for the DFA band (`(N_SURR_DFA)`). The methodology describes a single permutation throughout, and the ACF figure caption claims "mean over 10 realisations", which is wrong on both counts. The DFA caption is correct.
- **PCA dimensions.** " $K_{\text{show}} = 50$ components for eigenvalue analysis" and $\mathbf{P} \in \mathbb{R}^{M \times K_{\text{show}}}$ describe nothing in the code: `(fit_pca)` retains all 300 eigenvalues but only

K eigenvectors, K entries of `(explained_variance_ratio)`, and an $M \times K$ projection matrix, with $K = 20$.

- α_{FGN} in Eq. `(fgn_dfa_match)` appears exactly once in the paper and conflates the generator exponent α_0 with the measured exponent α ; the pipeline section states the relationship correctly.

All four are now corrected. The methodology states the three conventions explicitly — the burstiness spectrum and the DFA exponents averaged over N_{surr} permutations with mean $\pm 2\sigma$ bands, the autocorrelation reported for a single realisation because the shuffled ACF is indistinguishable from zero at every lag — and the erroneous "mean over 10 realisations" is gone. The one surviving occurrence of that phrase is in the DFA caption, where `(N_SURR_DFA = 10)` makes it correct.

Still stale, inside commented-out source that the authors intend to keep: the shuffle block states $N_{\text{surr}} = 10$ against `(n_surr = 20)` in `(config.py)`, and contains a sentence missing its first half. Neither is typeset, but both should carry a `(% STALE)` marker so they are not restored unexamined. No such marker is in the source yet.

3.10 Editorial changes (status: done)

- §1.2 retitled *The Altmann et al. mechanism* → *From word-level burstiness to topical organisation*. The mechanism is joint work and the old title over-attributed it; the citation remains in the first sentence.
- "biological variation" → "geographical distribution" in the list of example topics, so the examples match the semantic poles actually found (Darwin's PC2 is geographical).
- $K = 20$ stated once, where the burstiness spectrum is defined. The accompanying claim about higher-order components was initially written without evidence; it has since been measured and restated with figures (§3.8), and is a candidate for the robustness section of the Results rather than the introduction.
- Redundant restatements of "PCA maximises variance, not burstiness" and of the word-shuffled null's properties compressed.
- §Logical role of the null models now opens with a table (`(tab:null_taxonomy)`) setting out which statistical ingredient each sequence retains, and closes with the joint conclusion that both answers are negative and non-redundant. Given §3.6, the embedding geometry row reads *exact* for both null models, so the two differ in exactly one ingredient.
- `(rem:fgn_entropy)` concerns rank proximity versus semantic proximity and mentions no entropy; the label should be renamed while it is cited in only one place.

4. v11 → v12 — typesetting and the Supplementary Material

No numbers changed in this step. It converts the manuscript to the journal's format and splits it in two, to meet the Physica A limit of 30 typeset pages. The starting point was 44 pages with the old preamble and 37 pages under bare `(elsarticle)`.

4.1 Document class and references (status: done)

`\documentclass[preprint,authoryear]{elsarticle}` → `[preprint]`, and `\bibliographystyle{apalike}` → `elsarticle-num`. Physica A uses numbered references exclusively; `authoryear` would have been rejected at production. The explicit `geometry` and `natbib` loads were dropped, the first because it conflicts with the class defaults, the second because `elsarticle` loads `natbib` itself.

The class change takes effect; the `\bibliographystyle` change does not, for the reason set out in §4.6.

4.2 Split into main text + SI (status: done)

Eleven floats moved to `SI.tex`, which numbers its own floats S1, S2, ... via `\renewcommand{\thefigure}{S\arabic{figure}}` and the analogous redefinitions for tables, equations and sections.

moved to SI	why
per-text IET distributions (3 figures)	corroborate the <i>War and Peace</i> example kept in the main text
quantile sweep on the four texts	the corpus-level sweep, which is the probative one, stays in the main text
distribution of burstiness ratios	summary statistics are quoted in the text
mean ACF, corpus ACF decay, fraction significant, $\rho(1)$ distribution	four figures supporting one paragraph
ACF summary table, DFA exponent table	full numerical tables; conclusions are in the text

Cross-references run in one direction only. The main text carries `\usepackage{xr}` and `\externaldocument{SI}`, so `\ref` into the Supplementary Material resolves automatically and prints "Figure S5" without any edit to the body text. The SI never uses `\ref` into the main text — its few backward references are written in words. The alternative, `xr` in both files, is circular and needs four alternating compilations to converge; it was rejected as too fragile for submission. **Compile `SI.tex` first, then `paper3_main_v12.tex` twice.**

The SI uses `[H]` placement (package `float`) for every float, not `[htbp]`. A supplementary document has almost no running text — ten lines of prose against two hundred lines of floats — so the standard algorithm collapses all section headings onto the first page and then flushes the float queue, which puts the figures of §S1 after the heading of §S3. `[H]` pins each float where it is written.

4.3 Three figures deleted rather than moved (status: done)

The `(02_spectra_comparison)` figures for *Origin of Species*, *Great Expectations* and *Ulysses* were removed from the paper entirely. By §3.6 the three curves they plot are identical by construction, so the figures verify the construction rather than reporting a result; doing that once, in the *War and Peace* figure whose caption now says so explicitly, is enough. The one substantive sentence in those captions — the more gradual eigenvalue decay of *Ulysses* reflecting the diversity of its eighteen episodes — was moved into the body text.

`(tab:dfa_proj)` had been copied into the SI without being deleted from the main text, so it was set twice under a duplicated label; the main-text copy was removed.

4.4 Two sections were lost to a copy-paste error and recovered (status: done)

While the floats were being moved, a contiguous block of 658 lines was deleted from the main text by accident. It removed:

- §*Corpus-level generalisation in full* — *Corpus description, Component-resolved burstiness holds corpus-wide, Robustness to the threshold quantile, Distribution of burstiness ratios*, and the corpus table `(tab:corpus)` listing the 16 texts;
- §*Temporal persistence in semantic projection space in full* — *Conditional entropy of the embedding trajectory, A proxy measure: autocorrelation of projections, Empirical results: ACF of projections, Corpus-level ACF results, DFA of projected series*;
- four floats that existed in neither file afterwards: `(tab:corpus)`, `(fig:corpus_spectrum)`, `(fig:corpus_robustness)`, `(fig:acf_decay)`.

The cut began inside the *Robustness* subsection rather than at a section boundary, and ended in the middle of the DFA subsection, leaving `(fig:dfa_spectrum)` and `(fig:dfa_loglog)` behind. It joined in the middle of three blank lines, so **the file still compiled without a single error**. What exposed it was the label audit: four `(\ref)` pointing at labels present in neither document (`(sec:corpus)`, `(sec:entropy)`, `(tab:corpus)`, `(sec:corpus_robustness)`).

The block was restored from the previous revision and verified byte-for-byte against it. One float inside the deleted range was deliberately *not* restored: `(tab:dfa_proj)`, which had already been moved to the SI and would have come back as a duplicate.

Two lessons worth carrying into the remaining edits. First, a LaTeX manuscript that compiles is not a manuscript that is intact; only a label and reference audit catches this class of loss. Second, the audit found it in seconds — it is worth running after every structural edit, not only when something looks wrong.

4.5 `(tab:single_text_summary)` reformatted (status: done)

The table overflowed the text block. It was transposed — nine quantities down the rows, four texts across the columns — which also let the row labels carry the quantity names in words instead of bare symbols. The $B_{\max}$ entry was split into $B_{\max}$ and $k_{\max}$, so the

component attaining the maximum is a comparable number rather than a parenthesis inside a cell. The significance legend moved from a `\multicolumn` row after `\bottomrule` — where booktabs should have nothing — into the caption. No column was lost.

The table now shows the discrepancy discussed in §3.4 in a form a reader cannot miss: *Origin of Species* is the only text with no mark on r (0.295) but a mark on ρ (0.460*). This needs a sentence in the text, or it becomes the first question at review.

4.6 The reference list is still not in Physica A format (status: open)

`\bibliographystyle{apalike}` was changed to `\elsarticle-num` (§4.1), but that change has **no effect**. The bibliography is written by hand as a `\thebibliography` environment; `\bibliographystyle` selects a `.bst` for BibTeX to generate a `.bbl`, and BibTeX never runs here. The eleven entries are still in author-year form:

```
latex
```

```
\bibitem[Alvarez-Lacalle et al.(2006)]{AlvarezLacalle2006}
Alvarez-Lacalle, E., Dorow, B., Eckmann, J.-P., and Moses, E. (2006).
\newblock Hierarchical structures induce long-range dynamical correlations...
```

Every one of the eleven `\bibitem` commands carries an author-year optional argument, and the entry bodies follow apalike's field order. Physica A uses numbered references exclusively, so the list has to be rebuilt, not relabelled.

Since the bibliography needs expanding anyway — eleven references is thin for this journal — the efficient route is to build a real `refs.bib` now and let BibTeX produce the `.bbl` with `\elsarticle-num`. Doing it in the other order means formatting the new entries by hand twice.

The body uses `\citep` and `\citet` (18 calls). Those come from natbib, which `\elsarticle` loads itself, so they survive the switch to numbered mode — but `\citet` renders the author name from the `\bibitem` optional argument, which disappears when BibTeX takes over. Check those ten `\citet` calls read correctly after the change.

One entry, `\MontemurroDegliEsposti2025`, is in the bibliography and cited nowhere. Either cite it or drop it before submission.

5. Open items

Ordered by consequence, not by effort. Re-verified against the v12 source on 19 Sep; the four items that were on this list and are now applied have been removed rather than marked done.

1. **Write the Discussion.** §*Discussion and Conclusions* is still a commented-out `\TODO` block listing key points. It is the only section of the paper with no text in it, and the only item on this list that a referee would treat as disqualifying.

2. **Rebuild the bibliography as a `.bib` and expand it** (§4.6). The reference list is not in the journal's format and changing `\bibliographystyle` did not change it. Eleven references is also thin given how much of the framing rests on Altmann et al. (2012) and Alvarez-Lacalle et al. (2006). Both problems are cheapest to fix in one pass.
3. Recompute the filtered-rank DFA of §3.7 with the estimator used inside `fgn_surrogate` before quoting those numbers in the paper. The four columns as computed disagree with the pipeline's own `alpha_text` by up to 0.045, and of opposite sign for *Ulysses*.
4. State the fixed seeds in the methodology: `pca_seed=42`, `fgn_seed=42`, `n_surr=20` with seeds 0–19, 10^5 permutations with seed 0. The realisation *conventions* are now stated (§3.9); the seeds are not, and appear only in commented-out source.
5. Mark the commented-out source blocks the authors intend to keep with `% STALE`: the shuffle block still states $N_{\text{surr}} = 10$ against `n_surr = 20` in `config.py`, the $R = 150/200$ degradation sentence survives in a comment, and the conflated α_{FGN} of Eq. `fgn_dfa_match` survives in another. None is typeset; none should be restored unexamined either. No marker is in the source yet.
6. `paper_numbers.py` sections 2–5 still carry hardcoded "(old ...)" strings from v8. The `.tex` itself is correct and in sync; only the script's comparison column is stale, so its output for those sections currently flags differences that do not exist.
7. Typo: `(Remark~\ref{rem:fgn_cov},)` in the PCA-and-projection subsection has a stray comma inside the parenthesis.
8. Page count under `elsarticle` has not been measured since the split and the recovery. The limit is 30.

Applied since the 18 Sep list, and therefore no longer here: the four remaining $\textcircled{p} \rightarrow \textcircled{P}$ and the three symbol renames; the permutation marks and the Spearman column in `tab:single_text_summary`; the rewrite of the *Origin of Species* subsection around the PC6 maximum, including the sentence on the Pearson/Spearman discrepancy; the three stale covariance passages; the OOV rate, the K_{show} passage and α_{FGN} ; the placement of the $K = 50$ result in the robustness subsection; and the realisation conventions in the methodology.

6. Reproducing everything

```
bash

python scripts/run_corpus.py data/corpus          # corpus_*.csv
python scripts/paper_numbers.py data/corpus        # results/paper_numbers.md
python scripts/run_text.py data/corpus/<text>       # per-text figures
```

`paper_numbers.py` asserts that the permutation implementation reproduces `scipy.stats.pearsonr` to 10^{-9} on every text, so a silent divergence between the two code

paths cannot go unnoticed.

Permutation parameters are fixed: 10^5 pairings, seed 0. Both belong in the methodology section alongside the test itself.

Section 6 of `results/paper_numbers.md` now carries the verifications introduced above: M against M_{FGN} , multiset identity at three values of R , the filtered-rank DFA exponents, and the $K = 50$ burstiness refit. Each carries its own guard — a calibration column for the DFA estimator, an assertion that the refit reproduces the stored projections, and an assertion on the number of components returned, so that a silent truncation cannot be mistaken for a result.